Jay Doshi

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Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Relevant Coursework: Statistics, Kalman Filtering, Database Systems, Operating Systems, Software Engineering

Professional Projects

SportsSight Analytics | Python, OpenCV, YOLOv8, K-Means, NumPy, Matplotlib **August 2025 – December 2025**

- Fine-tuned YOLOv8 on a Roboflow football dataset to detect and track players and ball, reaching 90% detection accuracy.
- Implemented optical flow and perspective transformation in OpenCV to compute real-world player speed and distance in meters.
- Applied K-Means pixel clustering to segment players by t-shirt color for team assignment, raising segmentation accuracy to 75%.
- Generated real-time match analytics on ball possession and movement patterns from processed video frames.

Cloud-Native Load Balancer System | Docker, Linux, HAProxy, TCP/IP, VirtualBox, Apache **August 2025 – December 2025**

- Architected a three-node Linux LAMP system with primary, backup, and HAProxy servers, serving 500+ users with zero downtime.
- Engineered HAProxy Layer 4/7 routing and per-IP rate limiting, capping traffic at 200 req/sec while blocking abusive bursts.
- Automated rsync and database export/restore pipelines between nodes, cutting recovery time to under 5 minutes.
- Implemented encrypted Tailscale mesh networking between nodes to eliminate public exposure and enforce authenticated access.

DeepTruth-AI | Python, PyTorch, DistilBERT, Linux, Gemini API, GCP, AWS, React, MongoDB **January 2025 – March 2025**

- Fine-tuned DistilBERT on 9.5K+ labeled records, enabling dual-model scoring with 70% Gemini and 30% BERT weighting.
- Automated CI/CD deployment via GitHub Actions and GCP VMs, reducing manual release steps across the software lifecycle.
- Designed AWS Lambda and CloudWatch cron workflows for scheduled Transformer retraining and ML lifecycle management.
- Won RocketHacks 2025 for developing a scalable AI fact-checking product used by 1,000+ users over a 6-month period.

Smart Obstacle Avoidance Vehicle | C++, Embedded Systems, Arduino UNO, TinkerCAD **Github Link**

- Developed embedded C++ software reading ultrasonic sensors and driving servo motors for real-time obstacle detection.
- Built speed and distance tracking with an IR sensor and rotating grid plate, achieving 85% measurement accuracy.
- Integrated Bluetooth controls, reducing command response latency by ~60% and enabling near real-time mobile operation.
- Designed and tested a modular architecture with an L293D motor driver, ensuring reliable integration and smooth motor control.

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Integrate server infrastructure, VMware virtual environments, and backup systems to keep plant applications available.
- Monitor network and server health in SolarWinds NPM, flagging performance degradation before it escalates into outages.
- Collaborate with infrastructure, OT, and process-control engineers to validate system integrations and production cutovers.

Process Controls Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Engineered a classification-based anomaly detection tool on live gauge sensor data, reducing defective plate escapes by 30%.
- Developed ETL pipelines moving live sensor data from legacy OpenVMS Alpha into SQL Server, cutting manual effort by 40%.
- Shipped VB.NET journaling microservices to a clustered QMS app, restoring TCP/IP data flow from PLCs to SQL Server.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Imaged and deployed 150+ Windows devices using CLI tools for BIOS updates, data backups, and software installation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Increased control accuracy by 60% by analyzing flight-test signals from 15+ hardware trials in Python, NumPy, and Matplotlib.
- Prototyped PID control designs in MATLAB, trialing gain parameters for the PX4 drone flight controller.
- Decreased target navigation error by 40% by tuning a custom PID-based flight model in a ROS2-Gazebo digital twin.
- Scripted Linux CLI tools for data collection, system configuration, and test logging across hardware and simulation runs.

Technical Skills

Programming Languages & Frameworks: Python, C++, C, MATLAB, Java, JavaScript, SQL, Bash Scripting, React, Django, FastAPI

Computer Vision & ML: Computer Vision, OpenCV, YOLO, PyTorch, Deep Learning, Supervised/Unsupervised Learning, Fine-Tuning

Data & Visualization: NumPy, Pandas, Matplotlib, Power BI Dashboards, Tableau

Containers, OS & Cloud: Docker, Kubernetes, Linux, Windows, VMware, AWS (Lambda, S3, EC2), Google Cloud Platform

Developer Tools: Git, GitHub Actions, Linux CLI, Windows PowerShell, WSL, VS Code, Jupyter Notebook, ROS2, Gazebo, Claude-Code